BMC Medical Education

RESEARCH

Open Access

Application and effectiveness analysis of AI Intelligent tutoring system combined with knowledge graph in basic medical pathology teaching

Yu Yuanyuan[1] and Yuan Ying[1*]

Abstract

Background Basic medical pathology teaching suffers from fragmented knowledge and insufficient personalized learning, which impairs students' systematic knowledge construction and learning efficiency. This study aimed to apply an AI and knowledge graph integrated teaching model to address these challenges and provide references for basic medical education reform.

Methods An 18-week randomized controlled trial was conducted on 360 nursing students (180 each in experimental/control groups). The control group used traditional teaching, while the experimental group adopted the "AI + knowledge graph" integrated model with AI-assisted learning and knowledge graph reviews. SPSS was used for quantitative and qualitative data analysis.

Results The experimental group showed significantly better outcomes: final score (84.2 ± 6.9 vs. 74.1 ± 7.6), after adjusting for class-level clustering using a two-level mixed-effects model, adjusted mean difference = 8.2, 95% CI: 4.7–11.7, $P = 0.012$, Cohen's d = 1.13 (95% CI not computed), indicating a large effect size. knowledge mastery rate (90.3% vs.75.8%), clinical case analysis score (28.5±3.2 vs. 21.3±4.1, $P<0.001$). It also reduced ineffective learning time.

Conclusion This integrated model effectively promotes structured knowledge construction and learning efficiency, realizing the synergistic effect of AI and knowledge graph. It provides a replicable and practical reference for the digital and intelligent reform of basic medical education courses.

Keywords AI Intelligent Tutoring System, Knowledge Graph, Pathology Teaching, Basic Medical Education, Personalized Learning, Knowledge Construction

*Correspondence:
Yuan Ying
yyi-023@outlook.com
[1]Basic Department, Chongqing Medical and Pharmaceutical College, Chongqing, China

© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

Introduction

Currently, global medical education is undergoing a profound transformation centered on digitalization and intelligence. In China, this process has received policy support. The "14th Five-Year Plan for Health Human Resources Development" explicitly calls for promoting the deep integration of emerging technologies such as artificial intelligence (AI) and virtual reality (VR) with medical education, aiming to build a personalized and efficient teaching model. Against this backdrop, pathology, a core discipline bridging basic theory and clinical practice within the basic medical system, has become a key focus in the digital transformation of medical education. The quality of pathology instruction directly impacts the development of medical students' clinical thinking and disease diagnostic capabilities.

The knowledge system of pathology is highly complex, primarily for three reasons. First, it covers a broad range of topics, encompassing the etiology, pathological changes, and mechanisms of hundreds of diseases across various human organ systems. Second, its mechanisms are abstract and challenging, requiring students to understand dynamic processes at the cellular and molecular levels, such as the evolution of tumor atypia and inflammatory cascades. Third, the knowledge points are closely interrelated, with complex causal chains, as exemplified by the relationship between chronic hepatitis, cirrhosis, and hepatocellular carcinoma [1]. These characteristics pose significant challenges to the traditional teaching model: students often perceive knowledge as fragmented and struggle to construct a systematic cognitive framework, while instructors find it difficult to provide effective guidance tailored to individual differences due to rigid teaching schedules [2]. Survey data indicate that nearly 40% of students report difficulties in understanding, memorizing, and applying pathological knowledge, and nearly 30% of teachers believe traditional methods fail to meet personalized needs, highlighting the urgency of teaching reform.

To address these challenges, the academic community has begun exploring technology-enabled approaches, the development of which can be divided into three stages. In the early stage (2009–2022), efforts focused primarily on the internal optimization of pedagogy and curriculum systems, such as improving teaching effectiveness through curriculum integration or research mentorship programs, with limited deep integration of digital technologies [3, 4]. During the intermediate stage (2023–2024), with the rise of AI technology, pilot proposals and exploratory applications of single technologies emerged, such as discussions on the potential of AI in pathological image recognition and case-based teaching. However, these often lacked systematic implementation frameworks and structured handling of knowledge systems [5–7]. Recently (2025–present), research trends have shifted toward multi-technology integration. Scholars have begun emphasizing the integration of medicine and engineering and combining AI with knowledge visualization technologies to address the issue of knowledge fragmentation. However, specific collaborative mechanisms and scalable implementation models remain underdeveloped [8]. Multimedia technology has been effectively applied in the teaching of pathological cardiovascular system, verifying the value of technical tools in improving the teaching effect of specific pathological subsystems [9].

In summary, while existing research acknowledges the value of digital technologies, limitations persist, including relatively isolated technology applications, insufficient alignment with core curriculum needs, and a lack of empirical support for their effectiveness. In terms of knowledge graph construction technology, the accuracy of knowledge entity extraction is a key prerequisite for ensuring the quality of the graph [10]. The neural architectures for Named Entity Recognition proposed by Lample have been widely verified in medical text processing, providing a mature technical path for the extraction of pathological terms and knowledge points [11]. However, few studies have applied this technology to the construction of pathology-specific knowledge graphs, resulting in inconsistencies in the definition of core knowledge entities and affecting the effectiveness of subsequent knowledge association modeling. Therefore, exploring systematic knowledge extraction and representation methods is essential for advancing intelligent pathology education. Knowledge discovery systems provide fundamental support for identifying implicit and explicit knowledge from databases, laying a technical foundation for knowledge structuring and intelligent tutoring applications [12].

The proposal for this research topic is based on the alignment of three key aspects. First, at the needs level, the urgent demands of pathology education for knowledge structuring and personalized learning closely match the adaptive recommendation capabilities of AI systems and the structured representation abilities of knowledge graphs. Second, at the technology level, AI algorithms such as collaborative filtering and deep learning can already enable accurate analysis of learning behaviors and content delivery, while knowledge graph technologies based on RDF and graph databases can efficiently construct and maintain complex networks of medical knowledge associations. This combination of technologies has been empirically validated in the field of medical education [13]. Finally, cross-domain experience offers valuable insights: incremental update algorithms for knowledge graphs from computer science can address the rapid evolution of medical knowledge [14]; The Kirkpatrick

Four-Level Model from educational evaluation helps build a multidimensional assessment system beyond exam scores; and Self-Determination Theory from psychology provides a theoretical basis for designing motivating learning systems [15].

In light of this, the present study aims to develop and validate a teaching model that deeply integrates an AI Intelligent Tutoring System with a Knowledge Graph. This model is not merely a superficial combination of technologies; rather, it employs the knowledge graph as a structured knowledge engine and the AI system as a personalized learning agent. It seeks to fundamentally address the core challenges in pathology education—often characterized by high volume, complexity, and learning difficulties—thereby providing an innovative and practical exemplar for the digital reform of medical education.

Although this study focuses on pathology, the proposed bidirectional AI–knowledge graph integration framework is designed to be adaptable to other medical courses. The generalizability and implementation challenges across different knowledge domains warrant separate investigation.

Materials and methods

This study adopted a randomized controlled trial (RCT) research design with a two-group parallel comparison (experimental group vs. control group). The single intervention variable was the integration of AI intelligent tutoring system and knowledge graph into pathology teaching, and all other teaching variables were strictly controlled to ensure the validity of the experimental results.

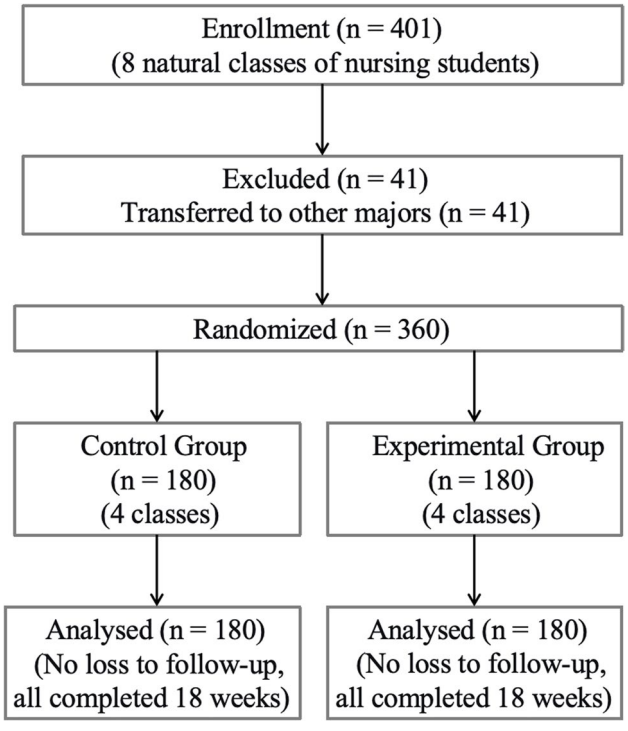


Fig. 1 Illustrates the participant flow throughout the trial according to the CONSORT statement

Selection and grouping of research subjects

A total of 401 first-year nursing students from eight natural classes at a medical college were initially enrolled. Later, 41 students transferred to other majors, resulting in 360 eligible participants. To ensure homogeneity and study validity, the following inclusion and exclusion criteria were established. The participant flow diagram is presented in Fig. 1.

Inclusion criteria:

- Enrolled in the 2024 nursing program at the college, having completed prerequisite courses (Human Anatomy and Physiology) with passing grades;
- No prior learning or training experience in basic pathological sciences;
- Voluntary participation and signed written informed consent;
- Possessing basic operational skills with electronic devices (smartphone or computer) to properly use the AI-powered intelligent tutoring system and the knowledge graph platform.

Exclusion criteria:

- Failure to complete or pass the prerequisite courses;
- Prior exposure to pathology content or participation in other pathology teaching reform experiments during the study period;
- Inability to complete the 18-week teaching experiment due to academic leave, major transfer, or clinical internship;
- Lack of basic electronic device operation skills, making it impossible to use the study's supporting teaching tools.

A cluster randomized controlled design was adopted, with randomization performed at the class level. Using SPSS 26.0 to generate random numbers, the eight classes were allocated to the experimental group (4 classes, 180 students) and the control group (4 classes, 180 students). Independent-samples t-tests showed no statistically significant differences in key baseline variables between the two groups ($P > 0.05$), and no significant differences were found among classes within each group (all $P > 0.05$). Baseline characteristics were as follows: sex distribution (experimental group: 78 males, 102 females; control group: 75 males, 105 females), mean prerequisite course score (experimental group: 71.2 ± 6.8; control group: 70.8

± 7.1), and average daily study time (experimental group: 38.5 ± 5.2 min; control group: 37.9 ± 5.5 min). These results confirmed balanced allocation and ruled out potential baseline confounding [16].

Because cluster randomization by class may introduce clustering effects (students within the same class share similar learning environments and peer influences), we calculated the intraclass correlation coefficient (ICC). Preliminary analysis indicated an *ICC* of 0.07 for the primary outcome, suggesting non-negligible clustering. Therefore, all primary and secondary outcomes were analyzed using a two-level mixed-effects model (level 1: individual student; level 2: class) with a random intercept. The model was fitted using the MIXED procedure in SPSS 26.0 with restricted maximum likelihood (REML) estimation [17].

This study was approved by the Ethical Review Committee of Chongqing Medical and Pharmaceutical College (Approval Number: KYLLSC20251201001). All research procedures were conducted in accordance with the ethical principles for medical education research in China. The trial was implemented from September 10, 2025 to January 14, 2026 (18 weeks, consistent with the semester cycle of the research). Written informed consent was obtained from all participants, who were fully informed of the research details and provided their voluntary consent in a standardized written format.

AI–knowledge graph integrated teaching system architecture

This system is driven by a dual-core mechanism combining a knowledge graph and an AI teaching assistant, forming a closed-loop intelligent teaching architecture characterized by "knowledge guiding learning and learning feeding back to knowledge."

The knowledge graph organizes pathology knowledge into a visualized, structured relational system, comprising 89 core knowledge points, 356 weighted relations, and hierarchical semantic associations. The graph is constructed based on the 10th edition of Pathology published by People's Medical Publishing House, following a dual-track approach of "top-down framework definition and bottom-up detail supplementation." It has been reviewed and validated by five senior pathology professors to ensure professional rigor.

The AI teaching assistant module analyzes students' learning status and provides personalized recommendations. It comprises three intelligent components: a knowledge-aware recommendation module, a pathology-adapted BERT QA module, and a dynamic path planning module [18, 19].

The knowledge graph supplies relation weights and semantic constraints to the AI, while the AI, in turn, feeds back students' high-frequency errors and concentrated queries to the graph, dynamically adjusting the relation weights among knowledge points. This enables continuous self-optimization of knowledge priorities.

All performance metrics reported above are based on internal tests using a separate cohort (2023 nursing students, non-overlapping with the RCT sample). No independent external validation has been performed. Although the test data are independent of the RCT cohort (avoiding direct circularity), the benchmarks remain internal, non-blinded, and without pre-registered thresholds. Optimistic bias is possible; external validation is needed.

Teaching implementation

The experimental cycle was 1 semester (18 weeks). The teaching resources of the two groups were strictly consistent (the same teaching team, the same theoretical and experimental teaching content, and unified assessment standards and scoring rules), with only differences in the teaching models. The specific implementation and quality control measures are as follows:

Control Group: Adopted the conventional model of "theoretical lectures + experimental teaching + paper assignments", including 48 class hours of theoretical lectures (explaining core knowledge points in the form of PPT combined with blackboard writing), 16 class hours of experimental courses (strengthening practical cognition through pathological section observation and virtual simulation software operation), and paper exercises assigned after class. Teachers mastered students' learning status through classroom questioning and correcting homework. The control group's daily learning duration was recorded via weekly learning logs and monthly questionnaires to maintain comparability.

Experimental Group : Based on the control group's teaching, independent learning with the AI intelligent tutoring system and knowledge graph review classes were added, and standardized operation requirements for the two types of auxiliary teaching were clearly defined: students needed to complete 1 weak-point diagnosis (through 5–10 targeted questions built into the system to automatically generate a knowledge point mastery rate report) and 1 set of associated exercises (exercises of weak knowledge points and strongly associated content pushed by the system based on diagnosis results, with 5–10 questions) daily, ensuring an average daily effective system usage duration of ≥ 35 min to avoid formal login.

A 45-minute weekly review session on knowledge graphs is conducted, following a fixed format: 10 min of instructor-led guidance, 25 min of group work, and a 10-minute summary and commentary. The instructor begins by outlining the core framework of the chapter and cross-chapter connections, integrating the knowledge graph. For example, the logical chain of

inflammation → repair → fibrosis → organ sclerosis. Students are then divided into groups of four to five to complete a task of mapping disease progression pathways. Finally, the instructor comments on gaps identified in each group's diagrams to reinforce the integrity of the knowledge system.This group-based mapping activity is designed to promote collaborative reflection and collaborative knowledge building, which, according to the Integrated Collaborative Reflection Model (ICRM), are key drivers for deep understanding and knowledge integration [20].

To avoid the impact of differences in teaching operation implementation on the validity of experimental results, multiple quality control mechanisms were set up: Two full-time research assistants formed a monitoring team. They weekly exported system usage logs of the experimental group students from the AI system backend to check compliance with the minimum effective operation requirement: ≥90% of students were required to complete daily weak-point diagnosis and related exercises with qualified effective duration. At the same time, the completion rate of group tasks in the knowledge graph review sessions was recorded on-site to ensure adherence to the standard process of path drawing and discussion.

For students who failed to meet the standards for two consecutive weeks, research assistants conducted one-on-one meetings with course teachers to analyze reasons for non-compliance (e.g., unskilled operation, time conflicts), provide system usage guidance, or adjust learning plans. For groups with low participation in knowledge graph tasks, teachers increased the frequency of guidance in subsequent courses to ensure the effective implementation of teaching intervention measures [21].

The basic data of the two groups, such as classroom attendance and homework submission rates, were compared monthly to ensure no significant differences. After the experiment, third-party statisticians conducted cross-validation of the teaching process data and score data of the two groups to exclude the impact of data recording errors on the result analysis.

Data collection and analysis

Quantitative data were collected using a scoring system combined with an AI backend platform, including baseline prerequisite course scores (Human Anatomy and Physiology, collected in Week 1 as a proxy for general academic ability, because students had no prior exposure to pathology), mid-term exam scores (Week 9), and final exam scores (Week 18) for both groups. Additional quantitative indicators for the experimental group comprised knowledge point mastery rate (calculated using Formula 1), learning time distribution, and the proportion of different question types in Q&A interactions.

Knowledge mastery rate

For each core knowledge point, the AI tutoring system automatically administered 5–10 quiz questions after the student completed the corresponding learning module. The mastery rate for a given knowledge point was calculated using the following formula:

$$K = \frac{C}{T} \times 100\%$$

where C represents the number of correct student responses and T is the total number of questions pertaining to the knowledge point. Mastery rates were updated daily based on the students' most recent responses. All data were recorded in the backend of the AI system.

Learning duration

For the experimental group, the AI system automatically recorded the total time spent on the platform each day, including time on micro-lecture videos, exercises and knowledge graph viewing. Effective learning time was defined as active engagement (e.g., answering questions, watching videos, navigating the knowledge graph), excluding idle time (no mouse or keyboard activity for more than 30 s) and repetition of content already mastered. For the control group, daily study time was collected through class attendance records and brief monthly surveys. Both groups' duration data were averaged over the 18 weeks period.

Upon completion of the experiment, two surveys were administered to the experimental group. The first was a 5-point Likert scale questionnaire consisting of 20 items across dimensions including system usability and the perceived helpfulness of the knowledge graph. This questionnaire was a brief, unvalidated satisfaction survey developed specifically for this study, without formal validity verification, reliability testing (Cronbach's α), factor analysis, or test-retest reliability. A total of 180 questionnaires were distributed, with 176 valid responses recovered. The second was a focus group interview, for which 12 students were selected representing three performance levels: high, average, and underachieving. Each interview lasted approximately 30 min to document participants' usage experiences and suggestions for improvement. The focus group interview protocol was developed specifically for this study, and its full English version is included as Supplementary Material. Qualitative analysis of the focus group transcripts followed thematic analysis procedures. Two researchers independently coded the first three transcripts using open coding, compared their coding schemes, and resolved disagreements through discussion to form an initial codebook. The remaining transcripts were coded independently by the two researchers, with inter-coder agreement of 87% (disagreements

resolved by consensus). Final themes were identified and refined through team discussion. Quotations presented in the results section represent typical or common views within each theme.

This study adopted a cluster randomized design with classes as clustering units, leading to non-independence of student observations. The primary outcome (final exam score) was analyzed using a two-level mixed-effects model: level 1 represented individual students, and level 2 included random intercepts for classes; the fixed effect was intervention group assignment. The model was implemented in SPSS 26.0 using the MIXED procedure with restricted maximum likelihood (REML) estimation. The intraclass correlation coefficient (ICC) was calculated from the null model (without covariates) to assess the magnitude of clustering. For sensitivity analysis, independent samples t-test ignoring clustering was also performed. A P-value < 0.05 was considered statistically significant.

Results

Teaching process data

The experimental group had an average of 2.6 daily logins and an average daily learning duration of 43.8 ± 6.5 min (including 42.3% for video learning, 35.1% for exercise practice, and 22.6% for knowledge graph viewing); the high-frequency access knowledge points of the knowledge graph were pathological characteristics of tumors(accounting for 31.2% of accesses) and pathogenesis of cardiovascular diseases (accounting for 25.7% of accesses), reflecting students' focus on complex knowledge points. During the experiment, the AI system answered a total of 6,842 questions for the experimental group, with the distribution of question types as follows: mechanism-related (e.g., differences between necrosis and apoptosis mechanisms) accounting for 48.5%, differentiation-related (e.g., pathological differentiation of various types of glomerulonephritis) accounting for 32.7%, and clinical association-related (e.g., association between pathological results and clinical symptoms) accounting for 18.8%, indicating that students pay more attention to in-depth understanding needs of knowing not only what but also why.

Learning effectiveness data

The average knowledge point mastery rate of the experimental group was 90.3%, 14.5% points higher than that of the control group (75.8%); among them, the difference in complex pathological mechanism knowledge points (such as metastasis pathways of malignant tumors and pathophysiological changes of liver cirrhosis) was the most significant, with the experimental group having a mastery rate of 87.6% and the control group only 65.2% ($P < 0.001$), reflecting the role of the knowledge graph in aiding the understanding of structured knowledge. The average daily effective learning duration of the experimental group (43.8 min) increased by 20% compared with the control group (36.5 min), but the proportion of repeated question brushing duration decreased by 16.8% (experimental group: 27.3% vs. control group: 44.1%), indicating that the personalized recommendation of the AI system effectively reduced ineffective learning behaviors.

As shown in Table 1, the two groups were comparable at baseline in terms of prerequisite course scores (71.2 ± 6.8 vs. 70.8 ± 7.1, mean difference = 0.4, 95% CI: -1.2 to 2.0, $P = 0.59$), confirming that any post-intervention differences cannot be attributed to baseline general academic ability.

The independent samples t-test without adjustment for clustering showed that the final exam score of the experimental group was significantly higher than that of the control group (84.2 ± 6.9 vs. 74.1 ± 7.6, MD = 10.1, $P < 0.001$). After adjusting for class-level random effects using the multilevel model, the.

Prerequisite course score: average score of Human Anatomy and Physiology. No pathology-specific pre-test was administered because students had no prior exposure to pathology.*.

between-group difference remained statistically significant (β = 8.2, 95% CI: 4.7–11.7, $P = 0.012$), with a Cohen's d of 1.13, indicating a meaningful practical effect despite

Table 1 Comparison of prerequisite course scores, mid-term exam scores, and final exam scores between the experimental group and control group

Assessment	Experimental Group ($n = 180$)	Control Group ($n = 180$)	Mean Difference (95% CI)	t / F / β	*P* value
Prerequisite course score (baseline)	71.2 ± 6.8	70.8 ± 7.1	0.4 (-1.2, 2.0)	t = 0.54*	0.59
Mid-term exam score	79.8 ± 7.0	71.5 ± 7.3	8.3 (6.5, 10.1)	t = 11.28*	< 0.001
Final exam score (unadjusted)	84.2 ± 6.9	74.1 ± 7.6	10.1 (7.9, 12.3)	t = 12.87*	< 0.001
Final exam score (adjusted for clustering)#	—	—	8.2 (4.7, 11.7)	β = 8.2**	0.012

*Notes: Data are presented as mean ± standard deviation

*Independent-samples t-test (ignoring clustering)

**Adjusted mean difference from two-level mixed-effects model with random intercept for class

#Intraclass correlation coefficient (ICC) for final exam score = 0.07 (95% CI: 0.02–0.14)

the modest adjustment for clustering. The *ICC* for final exam scores was 0.07 (95% CI: 0.02–0.14), indicating that the class clustering effect was not negligible [22].

To evaluate the potential impact of short-term non-adherence, two sensitivity analyses were performed. -Intention-to-treat analysis: All 180 students in the experimental group remained in their original groups.

-Adherence analysis: Ten students who failed to meet the criteria for two consecutive weeks were assigned to the low-adherence group, and the remaining 170 students formed the high-adherence group. The final examination scores of the two groups were compared. The results showed that the high-adherence group had a mean score of 84.4, compared with 81.6 in the low-adherence group, a difference of 2.8 points. The low-adherence group accounted for only 5.6% of the total experimental group, and all students ultimately received the complete intervention. Therefore, short-term non-adherence had a limited impact on the overall conclusions [23].

Feedback evaluation data

Students in the experimental group had a high recognition of the AI + knowledge graph model in a non-standardized, exploratory satisfaction survey developed for this study. The proportions of positive evaluations in key dimensions were: improved learning efficiency (91.5%), helped understand knowledge associations (88.1%), enhanced learning initiative (82.4%), and willing to use for a long time (85.2%); only 7.9% of students believed that the system's recommended content occasionally deviates from needs, and 5.1% believed that the knowledge graph update speed needs to be accelerated. A total of 176 valid questionnaires were recovered, and twelve students with different academic performances were selected for focus group interviews to collect subjective experience and improvement suggestions. These results should be interpreted as descriptive feedback rather than validated psychometric measures.(detailed qualitative findings see 4.1.4) [24].

Discussion

Horizontal comparison of own research results

Score dimension: leap from "basic mastery" to "in-depth understanding"

There was no significant difference in prerequisite course scores between the two groups (experimental: 71.2 ± 6.8, control: 70.8 ± 7.1; $P > 0.05$), confirming comparable baseline academic ability. Because students had not yet studied pathology, a pathology-specific pre-test was not feasible. The mid-term and final exam scores of the experimental group were significantly higher than those of the control group ($P < 0.001$). The mid-term exam score was 79.8 ± 7.0 in the experimental group versus 71.5 ± 7.3 in the control group (mean difference 8.3 points, $P < 0.001$). In the final examination, the adjusted score difference was 8.2 points (95% CI: 4.7–11.7), compared with an unadjusted difference of 10.1 points. From the perspective of score composition, the average score of the experimental group in subjective analysis questions (e.g., analyzing the pathological changes and clinical significance of a patient's liver puncture specimen) reached (28.5 ± 3.2) points (full score: 40 points), 33.8% higher than that of the control group (21.3 ± 4.1) points; while the objective memory questions (e.g., types of necrosis) only increased by 15.2% (Fig. 2). This difference shows that this model can not only assist students in strengthening the memory of knowledge points but also significantly improve their in-depth understanding abilities of knowledge application-clinical association-logical analysis, which is directly related to the structured association construction of the knowledge graph and the "accurate clinical case push" function of the AI system [25].

Learning behavior dimension: transformation from blind learning to accurate and efficient learning

The average daily effective learning duration of the experimental group increased by 20% compared to the control group, but the proportion of repeated question brushing decreased by 16.8%. The core reason lies in the personalized recommendation mechanism of the AI system: based on the knowledge point mastery rate calculation formula, the system can accurately identify students' knowledge blind spots (e.g., a student's mastery rate of pathological types of glomerulonephritis is only 52%) and prioritize pushing targeted learning resources, avoiding the blindness of brushing questions for the entire chapter in the control group. Simultaneously, the association path query function of the knowledge graph significantly improved the cross-chapter knowledge integration efficiency of students in the experimental group. For example, when learning about gastric cancer, students can quickly retrieve the pathological development path of chronic gastritis → intestinal metaplasia → gastric cancer and core mechanisms, reducing the time cost of fragmented memory of knowledge points. This is mutually confirmed by the high recognition rate of improved learning efficiency (91.5%) in the questionnaire survey [26](Fig. 3).

Feedback evaluation dimension: high subjective acceptance

In the experimental group, 85.2% of students responding to the non-standardized survey reported willingness to use the system long-term, suggesting positive subjective acceptance. However, given the exploratory nature of the survey instrument, these findings require confirmation with validated tools. First, the knowledge graph solved the pain point of disorganized knowledge points in traditional teaching, and 88.1% of students believed it helped

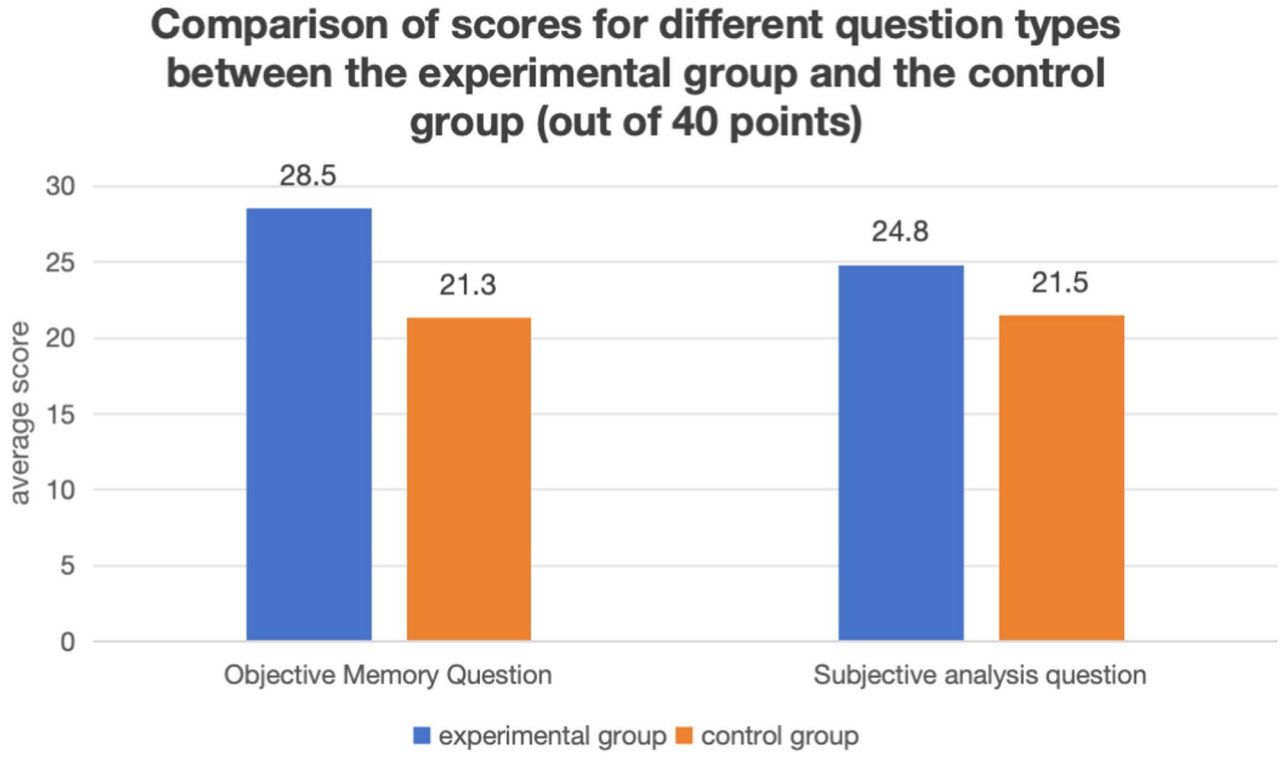


Fig. 2 Statistical chart of performance comparison between the experimental group and the control group

understand knowledge associations; second, the AI intelligent Q&A met the real-time learning needs, preventing students from affecting their learning progress due to accumulated questions. Eight interviewed students mentioned that when they encountered questions while doing exercises at midnight, the system could answer them immediately. They did not need to wait to ask the teacher the next day. However, attention should be paid to some negative feedback. For example, some students reported that the system's recommendations occasionally deviate from their needs. This may be due to the insufficient accuracy of the algorithm in identifying students' learning intentions. For instance, it may mistakenly classify extended reading needs as weak-point supplement needs. The function of learning intention labels needs to be introduced in subsequent optimizations.

Qualitative findings from feedback evaluation

Qualitative analysis of questionnaire and focus group interview data identified two core characteristics of students' subjective perception of the AI + knowledge graph model: high recognition of structural and personalized teaching value, and limited negative feedback with clear optimization directions.

First, the high-frequency positive feedback keywords reflected the model's solution to the core pain points of traditional pathology teaching. All twelve interviewed students mentioned accurately locating weak points, which verified the practical value of the AI system's personalized recommendation mechanism based on knowledge point mastery rate calculation—this function avoided blind learning and realized targeted breakthrough of knowledge blind spots, which was consistent with the quantitative result that the proportion of repeated question brushing in the experimental group decreased by 16.8%. Ten students pointed out that knowledge graph making cross-chapter connections clearer, which was due to the knowledge graph's structured association and visualization design, which constructed a systematic cognitive framework for students and solved the problem of fragmented knowledge in traditional teaching, matching the high mastery rate of complex pathological mechanism knowledge points (87.6%) in the experimental group. Eight students commented that intelligent Q&A saving time for asking teachers, which indicated that the pathology-adapted BERT model met students' real-time learning needs, making up for the defect of delayed Q&A in traditional teaching and effectively improving learning continuity.

Second, the limited negative feedback was mainly concentrated in technical optimization, and the students' demands were highly targeted. Only 7.9% of students reported "occasional deviation of recommended content from needs", which was attributed to the insufficient accuracy of the AI algorithm in identifying learning intentions—for example, the algorithm may misclassify

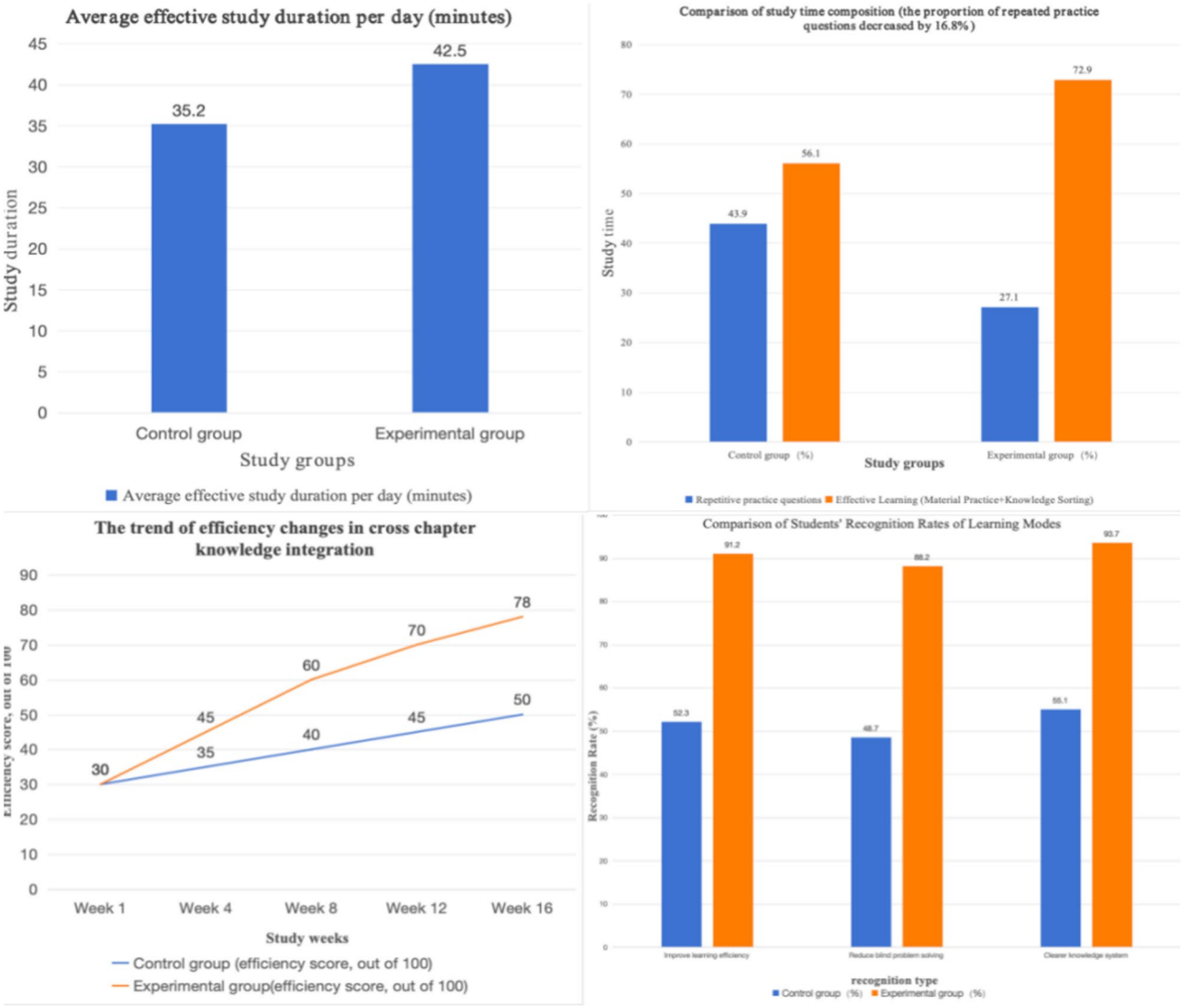


Fig. 3 Statistical chart of learning behavior comparison between the experimental group and the control group

students' extended reading needs as weak-point supplement needs, which suggested that the subsequent optimization of the system should introduce a "learning intention label" function to distinguish different learning demands. In addition, 5.1% of students hoped to "accelerate the update speed of the knowledge graph", and the interviewed students put forward two specific optimization suggestions: adding a linkage module between clinical cases and pathological knowledge (7 students) and optimizing the fluency of the mobile system interface (5 students). These suggestions were highly consistent with the disciplinary characteristics of pathology (close combination of basic theory and clinical practice) and the actual learning scenario of medical students (mobile learning demand), which provided clear practical directions for the iterative upgrade of the teaching model.

Third, the high long-term usage willingness (85.2%) reflected the students' active recognition of the model, which was a key sign of the transformation from passive acceptance to active learning in medical education. The core driving force of this active recognition was the "demand matching" between the model design and students' learning needs: the knowledge graph solved the structural confusion of knowledge, and the AI system met the personalized and real-time learning needs. This result also verified the rationality of the two-way collaboration design of the teaching model, and provided a subjective basis for the subsequent promotion and application of the model.

Vertical comparison with existing research: practical innovation, limitations and improvement directions

Towards a multidimensional assessment framework

Many current medical education studies tend to use examination scores as a key outcome measure. For example, Wang et al. evaluated the effectiveness of mind

map-based learning solely by final examination performance, showing a significant improvement in the intervention group's scores across all exam components [27]. In comparison, our study applied a multidimensional assessment framework that combines academic performance and learning behaviors to evaluate learning effectiveness in a more comprehensive manner, which may offer useful references for the reform of basic medical teaching.

Practical innovation from "laboratory pilot" to "real classroom"

Some related studies have been limited to small-sample sizes and short-term interventions under highly controlled conditions. As emphasized by Leppink et al., small samples yield low statistical power and a higher risk of unreliable results, which greatly restricts the generalizability of findings to real educational settings.This study overcame these limitations by conducting an 18-week full intervention in authentic teaching contexts with 360 students, and adopted three-level control to enhance the reliability and robustness of the results [28, 29].Similar to the practical application of knowledge graphs in perioperative nursing teaching [30], this study further verified the feasibility of knowledge graph technology in large-scale real medical teaching scenarios.

Consistency Control: The teaching teams, teaching content, and assessment standards of the experimental and control groups were kept completely consistent, with only changes in the teaching auxiliary models.

Discipline Adaptation Control: In view of the disciplinary characteristics of pathology (e.g., abstract mechanisms and close clinical associations), a three-dimensional association of pathological mechanism-clinical symptom-diagnostic standard was added to the knowledge graph, and a database of more than 500 real cases was embedded in the AI system to achieve accurate adaptation of technology to meet disciplinary needs.

User Willingness Control: The long-term usage willingness of students in the experimental group reached 85.2%, significantly higher than the average recognition rate of 72.6% under the "general technology application" model in similar international studies, providing a replicable practical plan for technology implementation in education.

Based on the above authentic teaching practice innovations, this study further proposes considerations for extending this model to other basic medical courses. Specific curriculum adaptation strategies are part of future implementation plans, as detailed in Sect. 5.3.

Research shortcomings and future improvement directions

The present study has several limitations that should be addressed in future research.

First, sample homogeneity and the limited number of clusters. All participants were nursing students recruited from a single medical college in Southwest China. This highly homogeneous sample limits the ability to capture differences in learning contexts across institutions of different tiers, academic disciplines, or teaching resources, thereby restricting the generalizability and external validity of the findings. Furthermore, this study employed a cluster-randomized controlled trial design with only four clusters (classes) per group, resulting in a limited number of clusters. Although a multilevel model was used to adjust for the clustering effect at the class level, the small number of clusters may still compromise the stability of variance estimates. Therefore, future studies should conduct multi-center, large-scale cluster-randomized trials—for example, by including two or three higher vocational medical colleges at different tiers and one undergraduate institution, with a total sample size of no less than 800 students—to validate the robustness of the results [31].

Second, systematic optimization for diverse educational settings. To enable large-scale application of the system, the association weights in the knowledge graph and AI recommendation algorithm need to be tailored to the teaching objectives of each institution. For example, in higher vocational education, the association between pathological manifestations and operational skills should be strengthened, whereas in undergraduate education, the linkage between pathological mechanisms and research advances should be deepened. Such adaptations would allow the construction of a stratified, adaptive teaching system capable of meeting the needs of different medical education scenarios.

Third, transferability potential and challenges to other basic medical courses. Although the current system was validated in the context of pathology, its architecture is domain-agnostic and can theoretically be transferred to other courses [32]. For instance, in pharmacology, the knowledge graph could be structured to build a pathology–disease–drug–therapeutic target relational network, facilitating understanding of drug mechanisms and adverse reaction pathways. However, such transfer may face two major challenges: first, the high cost of knowledge engineering, which requires deep involvement of domain experts and the establishment of a standardized ontology; second, the difficulty of course integration, given the substantial differences across courses in knowledge abstraction and the degree of clinical integration, which necessitates the design of flexible association weighting strategies.

Finally, we will develop and validate a standardized, psychometrically sound scale in future research to rigorously assess students' learning experiences and behavioral changes in AI- and knowledge graph-assisted teaching.

Potential biases

Teacher effect bias. Teachers were not blinded to group allocation, which may have introduced unconscious expectancy effects (e.g., differential attention or encouragement). Future studies should blind teachers or record interaction patterns to eliminate differences in teacher behavior or quantify the impact they may have [33].

Novelty effect. The experimental group's first exposure to the AI-KG system might have temporarily boosted engagement due to novelty rather than the intervention's inherent effectiveness. The 18-week duration partly mitigates this, but crossover designs or longer follow-ups are needed to separate novelty from true effects.

Hawthorne effect. The discrepancy in process monitoring intensity between the experimental and control groups may introduce Hawthorne effects and attention bias. In future studies, we will adopt an equal-intensity monitoring design or covert observation methods to mitigate the interference of such biases on the results.

Experimenter bias. The researchers were not blinded to group assignment and were involved in delivering the intervention and collecting outcome data, which may have unconsciously influenced student performance or data recording. Future studies should use blinded outcome assessors and, where possible, automated data collection to minimise this bias [34].

In summary, these biases may have led to overestimation of the intervention effect. Therefore, the findings should be interpreted with caution, and future studies are encouraged to employ more rigorous and appropriate methodologies to minimize the impact of these limitations on the results.

Absence of a pathology-specific pre-test. Because students had no prior exposure to pathology, a conventional pre-test on pathological content was not feasible. Therefore, we used prerequisite course scores to establish baseline equivalence in general learning ability, and evaluated the intervention effect solely by post-test comparison between groups.

Conclusions

Main findings summary

This study conducted an 18-week randomized controlled trial involving 360 nursing students in Southwest China. It explored the application effect of an AI intelligent tutoring system integrated with knowledge graphs in basic medical pathology teaching. The results showed that this integrated teaching model had significant effects. Compared with traditional teaching, the experimental group achieved a final mean score of 84.2 ± 6.9, which was significantly higher than the control group's 74.1 ± 7.6. After adjusting for class-level clustering effects, the mean difference was 8.2 points (95% CI: 4.7–11.7, $P = 0.012$), corresponding to an 11.1% relative improvement. The overall knowledge mastery rate reached 90.3%, an increase of 14.5% points. The experimental group exhibited a more significant improvement in accuracy for questions about complex pathological mechanisms and clinical case analyses. This indicated that the integrated model not only enhanced students' memory and mastery of basic pathological knowledge but also effectively improved their ability to apply theoretical knowledge to practical problems and develop clinical associative thinking.

Furthermore, the structured association characteristic of pathological knowledge graphs helped alleviate the issue of fragmented knowledge acquisition among students. The personalized recommendation function of the AI tutoring system mitigated the challenges of unfocused learning and inefficient review. Together, they formed a relatively complete closed-loop intelligent teaching framework. Among the experimental group, the proportion of effective learning time increased by 20% and the proportion of ineffective practice questions decreased by 16.8%. These results demonstrate the favorable performance of the integrated teaching model incorporating AI and knowledge graphs.

Nevertheless, due to the limited number of clusters (4 per group), the precision of the adjusted effect estimate is uncertain. Larger multi-cluster trials are needed to confirm these preliminary findings.

Practical implications

From the perspective of medical education practice, the research has important practical guiding significance for promoting the reform and optimization of basic medical pathology teaching. Firstly, it breaks through the limitations of traditional pathology teaching such as teacher-centered indoctrination and separation of theory and practice, provides a new feasible path for the digital and intelligent transformation of pathology teaching, and enriches the forms of intelligent teaching in basic medical education. Secondly, the integrated teaching model effectively solves the key difficulties in pathology teaching, such as abstract and complex knowledge, difficult logical association, and large individual differences in students' learning, helps students construct a systematic and structured pathological knowledge system, improves their learning efficiency and comprehensive quality, and provides effective support for cultivating high-quality nursing professionals with solid professional foundation and strong practical ability. Thirdly, the research verifies the feasibility and effectiveness of the combination of AI technology and knowledge graph in pathology teaching, provides practical reference for the popularization and application of similar intelligent teaching models in other basic medical courses (such as anatomy, physiology), and has important promotion value for promoting the high-quality development of basic medical education.

Future research recommendations

Based on the findings and limitations of this study, the following recommendations are proposed for future research.

Expand the research scope and participant populations

Future studies should validate the "AI + knowledge graph" teaching model across different grade levels, academic majors (e.g., clinical medicine, medical laboratory science), and types of medical institutions (e.g., undergraduate universities, vocational colleges) to further enhance the generalizability and adaptability of the model.

Optimize the functionality and performance of the teaching system

First, the dynamic updating mechanism of the pathology knowledge graph should be further refined by enriching the types of knowledge associations and expanding the repository of clinical case resources, thereby improving alignment with the latest textbooks and clinical practice. Second, the AI-driven recommendation algorithm and personalized learning path planning should be optimized by integrating a broader range of multidimensional learning behavior data, thereby enhancing the accuracy and relevance of intelligent tutoring. Third, longitudinal studies should be conducted to track the impact of the teaching model on students' subsequent academic learning and professional career development, providing deeper evidence for the long-term value of this model.

Strengthen the integration of the "AI + knowledge graph" teaching model with offline classroom instruction

A blended teaching model combining online intelligent tutoring with offline interactive classroom activities should be developed. This approach will further leverage the advantages of intelligent technologies to promote comprehensive improvements in the quality of pathology education.

Transfer the validated "AI + knowledge graph" integration framework to other basic medical science courses

For example: In pharmacology, the framework could highlight the logical chain of drug component–pathological mechanism–disease treatment. Specific nodes for drug targets and dynamic associations between drug efficacy and pathological changes could be added. In the recommendation module, priority should be given to matching drug mechanisms of action with pathological changes, while the question-answering module could be enhanced to explain the pathological mechanisms underlying adverse drug reactions. In physiology, the core framework could be structured around physiological function–organ structure–regulatory mechanism, with additional links connecting physiological abnormalities to pathological precursors and layered representations of complex physiological regulatory networks. The recommendation module could deliver animated demonstrations and simulated experiment resources tailored to students' weaknesses in understanding physiological mechanisms.

Abbreviations

AI	Artificial Intelligence
BERT	Bidirectional Encoder Representations from Transformers
RCT	Randomized Controlled Trial

Acknowledgements

We are particularly grateful to Dr. Hou Jiying and Dr. He Yuqin for their invaluable technical support in the construction of the AI intelligent tutoring system and pathology knowledge graph for this study.

Authors' contributions

Yu Yuanyuan: Conceptualization, Methodology, Investigation, Data curation, Funding acquisition, Writing – original draft, Writing – review & editing.Yuan Ying: Formal analysis, Investigation, Methodology, Writing – review & editing.

Funding

This work was supported by the 2024 Key Teaching and Research Project of Chongqing Medical and Pharmaceutical College (Grant No. CQYGZJG2404); the 2025 Project of Chongqing Municipal Education Commission (Grant No. Z2253462); the 2026 Project of Chongqing Society of Higher Education (Grant No. cqgj25253C); and the 2025 Project of Chongqing Educational Evaluation Research Association (Grant No. PJY2025152). The funders had no role in the study design, data collection and analysis, decision to publish, or preparation of the manuscript.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

This study was approved by the Ethical Review Committee of Chongqing Medical and Pharmaceutical College (Approval Number: KYLLSC20251201001). Participation in this study was voluntary, and informed consent was obtained from all participants.

Consent for publication

not applicable.

Competing interests

The authors declare no competing interests.

Received: 29 March 2026 / Accepted: 28 April 2026
Published online: 05 May 2026

References

1. Lin A, Xiong M, Tang B, et al. Decoding the hepatic fibrosis-hepatocellular carcinoma axis: from mechanisms to therapeutic opportunities. Hep Intl. 2025;19(4):732–59. https://doi.org/10.1007/s12072-025-10838-y.
2. Ohst A, Fondu BME, Glogger I, et al. Preparing learners with partly incorrect intuitive prior knowledge for learning. Front Psychol. 2014;5:1–11. https://doi.org/10.3389/fpsyg.2014.00664.
3. Zhang RQ, Yan AL, Su XL, et al. Exploration of Pathology-Centered Comprehensive Reform in Basic Medicine. Northwest Med Educ. 2009;17(5):975–7. doi:CNKI:SUN:XBYX.0.2009-05-060.
4. Pei F, Liu JY, Cui ZJ, et al. Experience of Tutorial System for Cultivating Innovative Undergraduates in Basic Medical Education Stage—A Case Study of Department of Pathology, Peking University Health Science Center. China High Med Educ. 2022;(2):14–5. doi:CNKI:SUN:ZOGU.0.2009-05-060.

5. Ma CH, Li Y. Discussion on Pathology Teaching Under the Background of Artificial Intelligence. Continuing Med Educ. 2025;39(3):130–3. doi:CNKI:SUN:JYJY.0.2025-03-033.
6. Gou W, Gao R. Exploration of Ideological and Political Teaching Model in Pathology Course Based on Case Analysis. Ability Wisdom. 2025;(25):37–40.
7. Myriam C, Megan M, Matthew MG, et al. Assessment of Pathology Learning Modules With Virtual Microscopy in a Preclinical Medical School Curriculum. Am J Clin Pathol. 2021;156(5):794–801. https://doi.org/10.1093/ajcp/aqab018.
8. Wei G, Hou XL, Shang TC. Exploration of Innovative Paths for Pathology Teaching Led by Modern Vocational Education. Educ Teach Forum. 2025;30:168–72. https://doi.org/10.20263/j.cnki.jyjxlt.2025.30.027.
9. Ma J, Liu YX, Fan LN, et al. Application of Informationized Multimedia Technology in the Teaching of Pathological Cardiovascular System. Chin Heart J. 2023;36(04):482–5.
10. Salih BA, Alotaibi S. A systematic literature review of knowledge graph construction and application in education. Heliyon. 2024;10(3):e25383. https://doi.org/10.1016/j.heliyon.2024.e25383.
11. Lample G, Ballesteros M, Subramanian S, et al. Neural Architectures for Named Entity Recognition. CoRR. arXiv. 2016;1603.01360.
12. Triantafyllou SA. Knowledge Discovery Systems: An Overview. In: Silhavy R, Silhavy P, editors. Software Engineering Methods in Systems and Network Systems. CoMeSySo 2023. *Lecture Notes in Networks Systems*. Cham: Springer; 2024. https://doi.org/10.1007/978-3-031-53549-9_16.
13. Chen Y. Evaluation of the impact of AI-driven personalized learning platform on medical students' learning performance. Comput Res Dev. 2017;54(12):2687–97. https://doi.org/10.3389/FMED.2025.1610012.
14. Chen DH, Yin SN, Le JJ. A Link Prediction Model for Clinical Temporal Knowledge Graph. Chin J Comput. 2023;46(06):1272–301. https://doi.org/10.7544/issn1000-1239.2017.20170640.
15. Cho Y, Carter S, Gullo C, et al. The GLASS Tool: A New Lens for Evaluating Generalist Leadership Competencies for Medical Education. Med Educ Online. 2025;30(1):2576571. https://doi.org/10.1080/10872981.2025.2576571.
16. Bolzern JE, Mitchell A, Torgerson DJ. Baseline testing in cluster randomised controlled trials: should this be done? BMC Med Res Methodol. 2019;19(1):1–5. https://doi.org/10.1186/s12874-019-0750-8.
17. Bell ML, Rabe BA. The mixed model for repeated measures for cluster randomized trials: a simulation study investigating bias and type I error with missing continuous data. Trials. 2020;21(1):148. https://doi.org/10.1186/s13063-020-4114-9.
18. Wang F, Zhu G, Zhu X. GKTP: A Personalized Knowledge Tracing and Predicting Method based on GNN. Procedia Comput Sci. 2025;266:1242–7. https://doi.org/10.1016/J.PROCS.2025.08.153.
19. Salam M, Farooq MS, Ikram A, et al. Integrated collaborative reflection model (ICRM) Part 1: role and process based conceptual study of collaborative reflection and collaborative knowledge building. Global Knowl Memory Communication. 2026. https://doi.org/10.1108/GKMC-03-2025-0242.
20. Ma D, Zhu H, Liao S, Chen Y, et al. Learning path recommendation with multi-behavior user modeling and cascading deep Q networks. Knowl Based Syst. 2024;294:111743. https://doi.org/10.1016/J.KNOSYS.2024.111743.
21. Deslauriers L, McCarty LS, Miller K, Callaghan K, et al. Measuring actual learning versus feeling of learning in response to being actively engaged in the classroom. Proc Natl Acad Sci U S A. 2019;116(39):19251–7. https://doi.org/10.1073/pnas.1821936116.
22. Hedges LV. Correcting a significance test for clustering. J Educational Behav Stat. 2007;32(2):151–79. https://doi.org/10.3102/1076998606298040.
23. White IR, Carpenter J, Horton NJ. Including all individuals is not enough: Lessons for intention-to-treat analysis. Clin Trials. 2012;9(4):396–407. https://doi.org/10.1177/1740774512450098.
24. Flake JK, Fried EI. Measurement schmeasurement: Questionable measurement practices and how to avoid them. Adv Methods Practices Psychol Sci. 2020;3(4):456–65. https://doi.org/10.1177/2515245920952393.
25. Li L, Wang P, Yan J, Wang, et al. Real-world data medical knowledge graph: Construction and applications. Artif Intell Med. 2020;103:101817. https://doi.org/10.1016/j.artmed.2020.101817.
26. Li A, Li Y, Gao X. Personalized Learning Path Recommendation Based on Knowledge Graphs. Surv Electronics. 2026;15(1):238–238. https://doi.org/10.3390/ELECTRONICS15010238.
27. Wang J, Li Q, Zhao L, et al. Using Mind Maps to Improve Medical Student Performance in a Pharmacology Course at Kunming Medical University. J Coll Physicians Surgeons–Pakistan: JCPSP. 2017;27(7):404–8.
28. Leppink J, Winston K, O'Sullivan P. Statistical significance does not imply a real effect. Perspect Med Educ. 2016;5(2):122–4. https://doi.org/10.1007/s40037-016-0256-6.
29. Ye Q, Wang S, Liu Y, et al. Application of AI-assisted multi-advisor system combined with BOPPPS teaching model in clinical pharmacy education. BMC Med Educ. 2025;25(1):783. https://doi.org/10.1186/S12909-025-07394-2.
30. Liu F, Shen M, Shen Y, et al. Clicking one dot opens a whole new world: a qualitative study on using knowledge graphs in surgical nursing education. BMC Med Educ. 2025;25(1):1615. https://doi.org/10.1186/S12909-025-08226-Z.
31. Fourie A, Anika Fourie, Ahtiala M, Black J, et al. Enhancing prone positioning and skin damage prevention education: A multicenter randomized controlled non-inferiority trial. J Tissue Viability. 2024;33(2):298–304. https://doi.org/10.1016/J.JTV.2024.02.008.
32. Liu X, Shen P, Xu X, et al. The study of blended teaching based on knowledge graph: the case of clinical microbiology examination course. Educ Inform Technol. 2025;30(16):22841–63. https://doi.org/10.1007/S10639-025-13626-1.
33. Trouilloud DO, Sarrazin PG, Martinek TJ, et al. The influence of teacher expectations on student achievement in physical education classes: Pygmalion revisited. Eur J Social Psychol. 2002;32(5):591–607. https://doi.org/10.1002/ejsp.109.
34. Asbjørn H, Skou TAS, Frida E, et al. Observer bias in randomised clinical trials with binary outcomes: systematic review of trials with both blinded and non-blinded outcome assessors. BMJ. 2012;344:e1119. https://doi.org/10.1136/bmj.e1119.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.